EE2019–Analog Systems and Lab: Tutorial 4

Nagendra Krishnapura, Shanthi Pavan

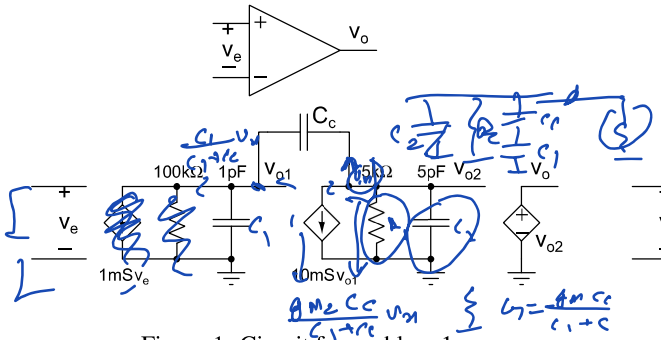


Figure 1: Circuit for problem 1

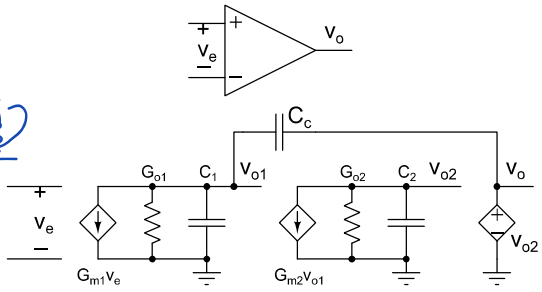


Figure 2: Circuit for problem 2

1. Fig. 1 shows the internal schematic of a Miller-compensated opamp. This opamp is used to realize a unity gain, non-inverting amplifier.

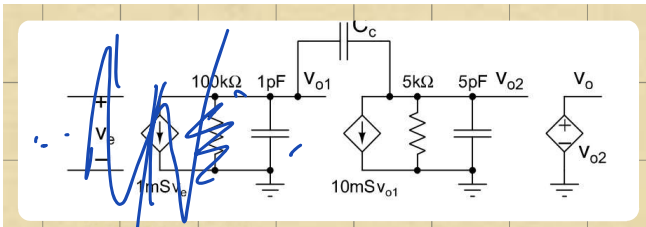
   - What is the phase margin?
   - Determine C_c so that the phase margin is 60°.
   - If the same opamp is used without any change to realize an inverting amplifier of gain -4, what are the phase margin and the closed loop bandwidth?
   - Re-design the opamp (value of C_c) so that when an inverting amplifier of gain -4 is realized using it, the phase margin is 60°. Compare the three cases wrt the following aspects: (a) Closed loop bandwidth, (b) phase margin, (c) phase lag contributed by the right-half-plane zero at the unity loop gain frequency.
   - Compare the bandwidths you obtain to the ones in the previous tutorial in which you simply increased C_1.

   While determining the unity loop gain frequency, phase margin, and C_c, do the calculations with and without the approximation $C_c \gg C_{1,2}$.

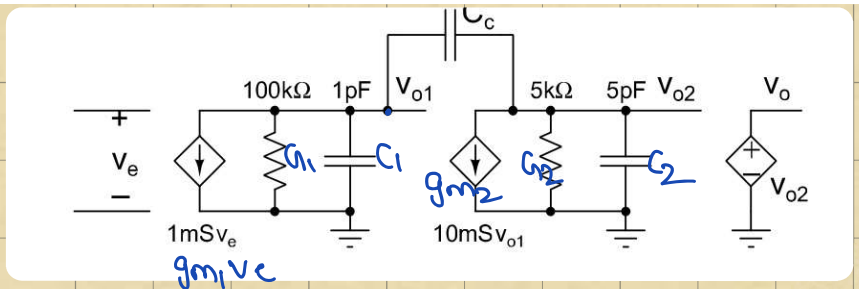
2. Determine the transfer function of the opamp in Fig. 2. How does it differ from the conventional Miller compensated opamp in the previous problem?

3. It is common to approximate the unity loop gain frequency as $\omega_{u,loop} \approx L_0 p_1$ where L_0 is the dc loop gain and p_1 is the dominant pole. If the loop gain is a second order function $L(s) = L_0/(1 + s/p_1)(1 + s/p_2)$, determine the exact unity loop gain frequency and the phase margin for the following cases: (a) $p_2 = 4L_0 p_1$, (b) $p_2 = 2L_0 p_1$, and (c) $p_2 = L_0 p_1$. Compare them to the values obtained using the approximation above. $L_0 \gg 1$.

   (This approximation is very commonly used for hand calculations, but you should know how much error you end up with while doing so.)



- What is the phase margin?
- Determine C_c so that the phase margin is 60°.
- If the same opamp is used without any change to realize an inverting amplifier of gain -4, what are the phase margin and the closed loop bandwidth?



$g_{m1}v_e$

$$V_{o1}(G_1+sC_1)+g_{m1}v_e+(V_{o1}-V_{o2})sC_c=0$$

$$V_{o2}(G_2+sC_2)+g_{m2}V_{o1}+(V_{o2}-V_{o1})sC_c=0$$

$$V_{o1}(G_1+sC_1+sC_c)+v_e(g_{m1})-V_{o2}(sC_c)=0$$

$$V_{o1}(g_{m2}-sC_c)+V_{o2}(G_2+sC_2+sC_c)=0$$

$$V_{o2}\frac{(G_2+sC_2+sC_c)(G_1+sC_1+sC_c)}{(sC_c-g_{m2})}+v_e(g_{m1})-V_{o2}(sC_c)=0$$

$$V_{o2}\frac{(s^2(C_c+C_2)(C_1+C_c)+s(C_2G_1+C_cG_1+G_1G_2+C_cG_2)+G_1G_2)}{sC_c-g_{m2}}-V_{o2}\frac{(s^2C_c^2-sC_cg_{m2})}{sC_c-g_{m2}}=-v_e(g_{m1})$$

$$=V_{o2}[s^2(C_1C_2+C_cC_1+C_cC_2)+s(C_2G_1+C_cG_1+C_1G_2+C_cG_2+C_cg_{m2})+G_1G_2]=-v_e(g_{m1})(sC_c-g_{m2})$$

$$H(s)=\frac{g_{m1}(g_{m2}-sC_c)}{G_1G_2+s[G_1C_2+G_1C_c+G_2G_1+G_2C_c+g_{m2}C_c]+s^2[C_1C_2+C_1C_c+C_2C_c]}$$

$$\frac{\frac{g_{m1}g_{m2}}{G_1G_2}\left(1-s/(g_{m2}/C_c)\right)}{\left\{1+s\left[C_2/G_2+C_c/G_2+C_c/G_1+1+\frac{g_{m2}C_c}{G_1G_2}\right]+s^2\left[\frac{C_1C_2+C_cC_1+C_cC_2}{G_1G_2}\right]\right\}}$$

$$p_1<<p_2 \Rightarrow \frac{1}{p_1}=\frac{G_1C_2+G_1C_c+G_2G_1+G_2C_c+g_{m2}C_c}{G_1G_2}$$

$$\Rightarrow p_1=\frac{G_1G_2}{G_1C_2+G_1C_c+G_2G_1+G_2C_c+g_{m2}C_c}$$

$$p_2 = \frac{G_1C_2 + G_1C_c + G_2G_1 + G_2C_c + g_mC_c}{C_1C_2 + C_cC_1 + C_2)}$$

$$\omega_u = A_0p_1 = \frac{g_{m_1}g_{m_2}}{G_1G_2}p_1$$

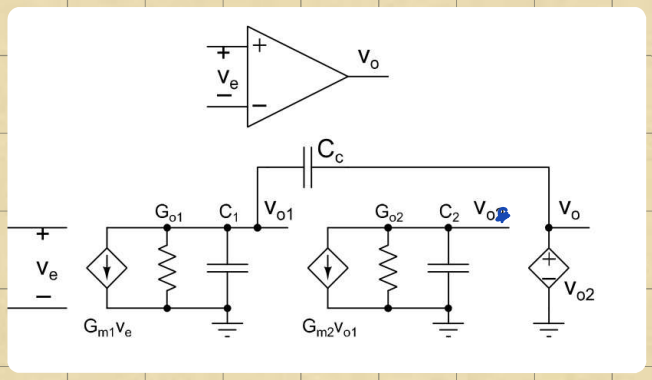
$$\phi = 180 - 90 - \tan^{-1}\left(\frac{\omega_u}{p_2}\right) + \tan^{-1}\left(\frac{\omega_u C_c}{g_{m_2}}\right)$$

$$\phi = \tan^{-1}\left(\frac{(G_1C_2 + G_1C_c + G_2G_1 + G_2C_c + g_mC_c)^2}{C_1C_2 + C_cC_1 + C_2)\, g_{m_1}g_{m_2}}\right) + \tan^{-1}\left(\frac{g_{m_1}C_c}{G_1C_2 + G_1C_c + G_2G_1 + G_2C_c + g_{m_2}C_c}\right) \to 0$$

$$\phi_m = \tan^{-1}\frac{10C_c^2}{6\times10^{12}C_c + 5\times10^{24}}$$

$$\phi_m = 60 \Rightarrow C_c = 1.59\ pf$$

$$K = -4 \Rightarrow \beta = \frac{1}{5} \Rightarrow \boxed{\omega_{ugf} = \frac{A_0p_1}{5}}$$



$$(V_{o1})(SC_1) + V_{o1}(G_{o1}) + G_{m1}V_e + (V_{o1} - V_o)(SC_c) = 0$$

$$G_{m2}V_{o1} + (G_{o2})(V_o) + (V_o)(SC_2) = 0$$

$$V_{o1}(SC_1 + G_{o1} + SC_c) = V_o(SC_c) - V_e(G_{m1})$$

$$V_{o1}(G_{m2}) = -V_o(SC_2 + G_{o2})$$

$$\Rightarrow \frac{SC_1 + G_{o1} + SC_c}{G_{m2}} = -\frac{SC_c}{SC_2 + G_{o2}} + \frac{V_e}{V_o}\frac{G_{m1}}{SC_2 + G_{o2}}$$

$$\Rightarrow \frac{\frac{G_{m1}}{SC_2 + G_{o2}}}{\frac{SC_1 + G_{o1} + SC_c}{G_{m2}} + \frac{SC_c}{SC_2 + G_{o2}}} = \frac{V_o}{V_e}$$

$$\Rightarrow \frac{G_{m1}G_{m2}}{(SC_1 + G_{o1} + SC_c)(SC_2 + G_{o2}) + SC_cG_{m2}} = H(S)$$

$$H(S) = \frac{G_{m1}G_{m2}}{S^2(C_1C_2 + C_2C_c) + S(C_1G_{o2} + C_cG_{o2} + C_2G_{o1} + C_cG_{m2}) + G_{o1}G_{o2}}$$

$$= \frac{G_{m1}G_{m2}/G_{o1}G_{o2}}{S^2\left[\frac{(C_1 + C_c)C_2}{G_{o1}G_{o2}}\right] + S\left[\frac{C_1 + C_c}{G_{o1}} + \frac{C_2}{G_{o2}} + \frac{C_cG_{m2}}{G_{o1}G_{o2}}\right] + 1}$$

$P_2 = -b/a \quad P_1 = c/b$

smaller pole: $\dfrac{-1}{\frac{C_1+C_c}{G_{o1}} + \frac{C_2}{G_{o2}} + \frac{C_c G_{m2}}{G_{o1}G_{o2}}}$

larger pole: $-\dfrac{G_{o2} + G_{o1}\left(\frac{C_2}{C_1+C_c}\right) + G_{m2}\left(\frac{C_c}{C_1+C_c}\right)}{C}$

3. It is common to approximate the unity loop gain frequency as $\omega_{u,loop} \approx L_0 p_1$ where L_0 is the dc loop gain and p_1 is the dominant pole. If the loop gain is a second order function $L(s) = L_0/(1+s/p_1)(1+s/p_2)$, determine the exact unity loop gain frequency and the phase margin for the following cases: (a) $p_2 = 4L_0p_1$, (b) $p_2 = 2L_0p_1$, and (c) $p_2 = L_0p_1$. Compare them to the values obtained using the approximation above. $L_0 \gg 1$.

(This approximation is very commonly used for hand calculations, but you should know how much error you end up with while doing so.)

$$|L(s)| = \frac{L_0}{\sqrt{(1+\omega_u^2/p_1^2)(1+\omega_u^2/p_2^2)}}$$

$$\Rightarrow \omega_u^4\left(\frac{1}{p_1^2p_2^2}\right) + \omega_u^2\left(\frac{1}{p_1^2}+\frac{1}{p_2^2}\right) + 1 = L_0^2$$

$$\Rightarrow \omega_u^4 + \omega_u^2(p_1^2+p_2^2) + (1-L_0^2)(p_1^2p_2^2) = 0$$

$$\omega_u^2 = \frac{-(p_1^2+p_2^2) + \sqrt{(p_1^2+p_2^2)^2 - 4(1-L_0^2)p_1^2p_2^2}}{2}$$

$$= -\frac{(p_1^2+p_2^2)}{2}\left[1 - \sqrt{1+4(L_0^2)\left(\frac{p_1p_2}{p_1^2+p_2^2}\right)^2}\right]$$

$p_2 = KL_0p_1 \Rightarrow$

$$\omega_u^2 = -\frac{(p_1^2+K^2L_0^2p_1^2)}{2}\left[1 - \sqrt{1+4L_0^2\left(\frac{KL_0p_1^2}{p_1^2+K^2L_0^2p_1^2}\right)^2}\right]$$

$$= \frac{p_1^2}{2}(1+K^2L_0^2)\left[\sqrt{1+\frac{4K^2L_0^4}{(1+K^2L_0^2)^2}} - 1\right]$$

$$\omega_u = p_1L_0\sqrt{\frac{1}{2}\left(\frac{1}{L_0^2}+K^2\right)\left[\sqrt{1+\frac{4K^2L_0^4}{(1+K^2L_0^2)^2}} - 1\right]}$$

since $L_0 \gg 1$ $\quad \omega_u = p_1L_0\sqrt{\frac{K^2}{2}\left[\sqrt{1+4/K^2} - 1\right]}$

$$= p_1L_0\sqrt{\frac{K}{2}\left(\sqrt{K^2+4} - K\right)}$$

a) $K=4 \Rightarrow \omega_u = p_1L_0\sqrt{2(2\sqrt{5}-4)} = 0.972\,p_1L_0$

b) $K=2 \Rightarrow \omega_u = p_1L_0\sqrt{2\sqrt{2}-2} = 0.910\,p_1L_0$

c) $K=1 \Rightarrow \omega_u = p_1L_0\sqrt{\frac{\sqrt{5}-1}{2}} = 0.786\,p_1L_0$

Problem 4

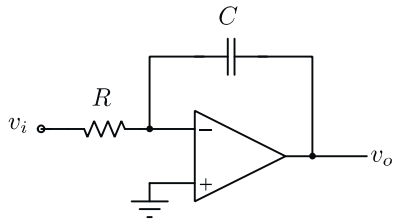


Figure 1: Circuit for Problem 1.

Fig. 1 shows an integrator. The opamp is ideal. The capacitor is initially uncharged. $v_i = \sin(\omega_o t)u(t)$, where $\omega_o = 1/RC$ and $u(t)$ is the unit step function. Draw to scale, on the same graph, v_i and v_o. Repeat with $v_i = \cos(\omega_o t)u(t)$.

Problem 5

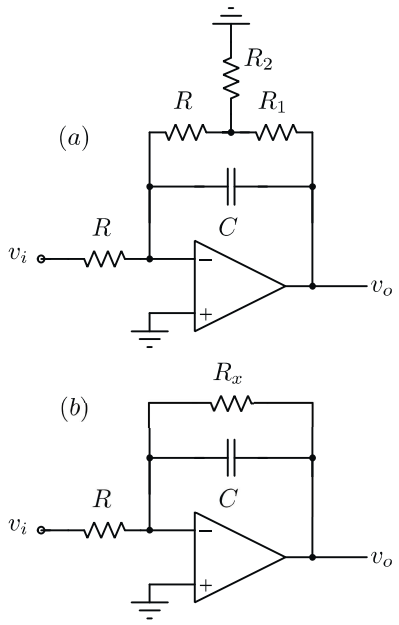


Figure 2: Circuit for Problem 2.

All opamps are ideal in Fig. 2. Determine the dc gain and 3-dB bandwidth of the circuit of Fig. 2(a). What R_x should be chosen in the circuit of Fig. 2(b) to obtain the same transfer function?

Evaluate R_x in the limiting case when $R_1, R_2 \ll R$. What might be the utility of the T-network in Fig. 2(a)?

Problem 6

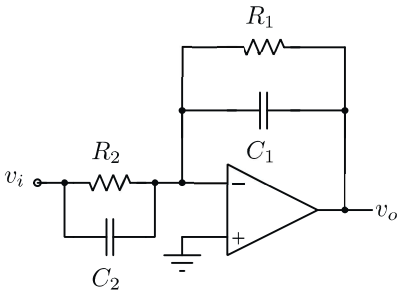


Figure 3: Circuit for Problem 3.

Determine the transfer function of the circuit of Fig. 3. Sketch a Bode plot.

Problem 7

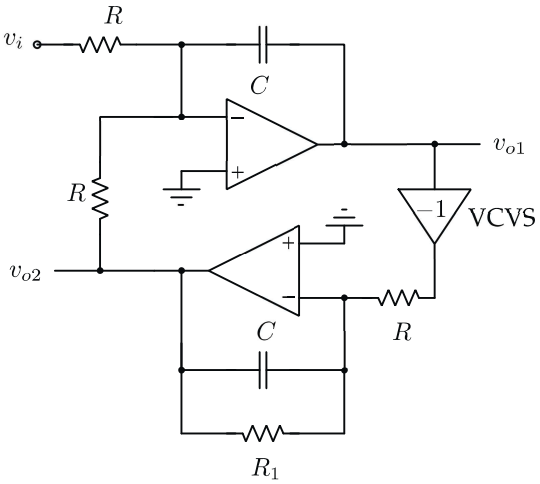


Figure 4: Circuit for Problem 4.

The opamps are ideal. Determine the transfer functions from the v_i to v_{o1} and v_{o2}.

Problem 8

The opamps are ideal. The initial conditions are marked. Plot the waveforms v_{o1} and v_{o2}.

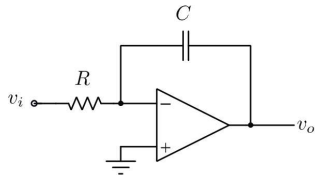


Figure 1: Circuit for Problem 1.

Fig. 1 shows an integrator. The opamp is ideal. The capacitor is initially uncharged. $v_i = \sin(\omega_o t)u(t)$, where $\omega_o = 1/RC$ and $u(t)$ is the unit step function. Draw to scale, on the same graph, v_i and v_o. Repeat with $v_i = \cos(\omega_o t)u(t)$.

$V_o R + \frac{V_i}{SC} = 0$ (@ steady state)

$V_o = -\frac{V_i}{S/(1/RC)}$ $\omega_o = 1/RC$

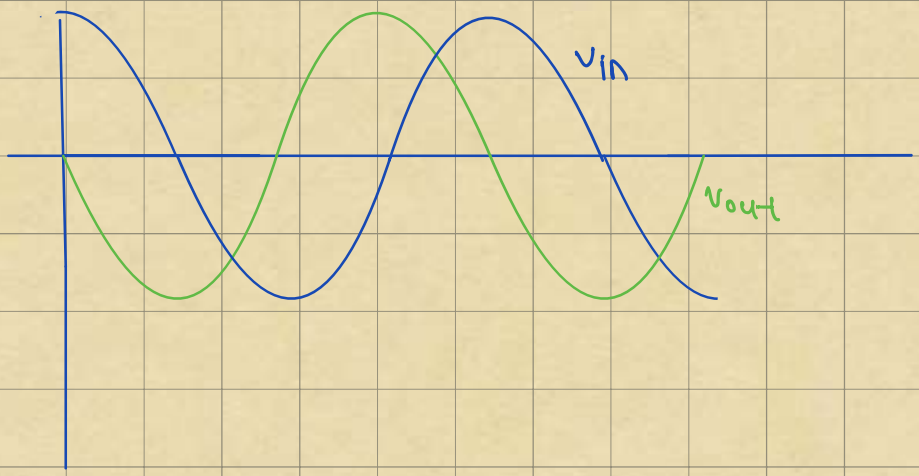
$V_i(S) = \frac{\omega_o}{(S^2+\omega_o^2)}$

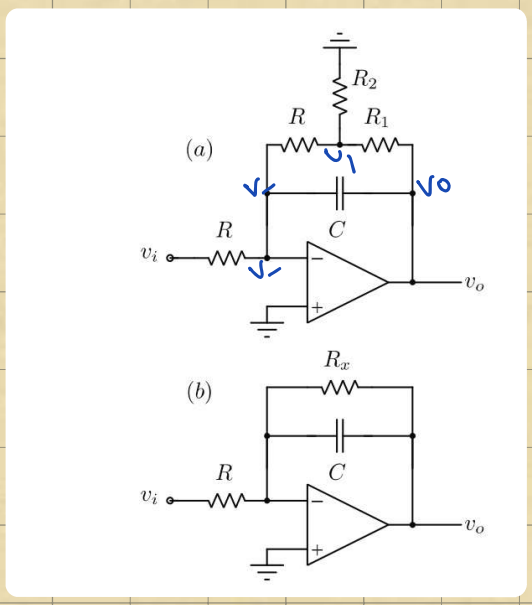
$\Rightarrow V_o(S) = \frac{-\omega_o^2}{S(S^2+\omega_o^2)} = S/S^2+\omega_o^2 - 1/S$

$v_o(t) = (\cos(\omega_o t) - 1)\,u(t) = -2\sin^2\frac{\omega_o t}{2}\,u(t)$



$V_i(S) = \frac{S}{S^2+\omega_o^2} \Rightarrow V_o(S) = -\frac{\omega_o}{S^2+\omega_o^2} \Rightarrow V_o(t) = -\sin(\omega_o t)\,u(t)$





All opamps are ideal in Fig. 2. Determine the dc gain and 3-dB bandwidth of the circuit of Fig. 2(a). What R_x should be chosen in the circuit of Fig. 2(b) to obtain the same transfer function?

Evaluate R_x in the limiting case when $R_1, R_2 \ll R$. What might be the utility of the T-network in Fig. 2(a)?

realize large feedback resistance with smaller resistances

Steady state $V_- = 0$

$$\frac{-V_i}{R} + \frac{-V_o}{1/sC} + \frac{-V_1}{R} = 0$$

$$\frac{V_1}{R_2} + \frac{V_1 - V_o}{R_1} + \frac{V_1}{R} = 0$$

$$\Rightarrow V_1 = \frac{V_o}{R_1}\left[\frac{1}{1/R_2 + 1/R + 1/R_1}\right]$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{-1}{sCR + \frac{1}{1 + R_1(1/R + 1/R_2)}}$$

DC gain: $s = 0 \Rightarrow -\left(1 + R_1\left(\frac{1}{R} + \frac{1}{R_2}\right)\right)$

$$|\omega_{3dB} \cdot RC| = \frac{1}{1 + R_1(1/R + 1/R_2)} \Rightarrow \omega_{3dB} = \frac{1/RC}{1 + R_1(1/R + 1/R_2)}$$

(b) $$\frac{V_o}{V_i} = \frac{-1}{R/R_x + sCR} \Rightarrow \frac{R}{R_x} = \frac{1}{1 + R_1(1/R + 1/R_2)}$$

$$\Rightarrow R_x = R\left(1 + R_1/R_2\right)$$

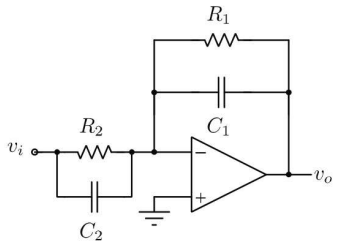


Figure 3: Circuit for Problem 3.

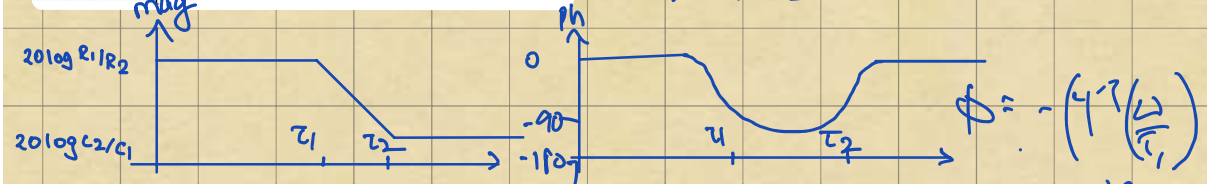
Determine the transfer function of the circuit of Fig. 3. ketch a Bode plot.

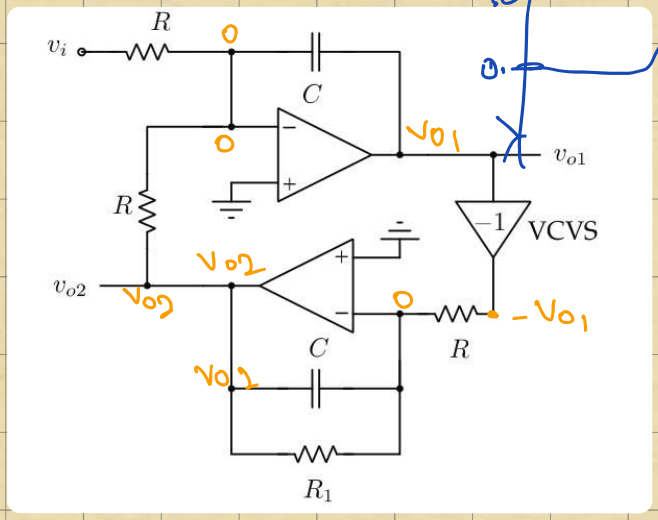
$$v_i \times \frac{R_2}{1 + sC_2R_2} = -v_o \times \frac{R_1}{1 + sC_1R_1}$$

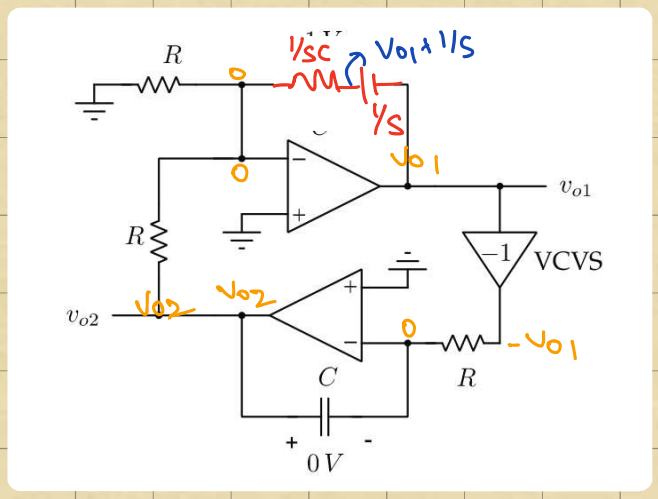
$$\Rightarrow V_o/V_i = -(R_1/R_2)\left(\frac{1 + sC_2R_2}{1 + sC_1R_1}\right)$$

$$= -(R_1/R_2)\left(\frac{1 + s/\tau_1}{1 + s/\tau_2}\right)$$

If: $\tau_1 < \tau_2$







$$\frac{-V_{o2}}{R} + \frac{-V_i}{R} + \frac{-V_{o1}}{1/sC} = 0$$

$$\frac{V_{o1}}{R} + \frac{-V_{o2}}{R_1/1+sCR_1} = 0$$

$$\Rightarrow V_{o1} = V_{o2}\left(\frac{R}{R_1}\right)(1+sCR_1)$$

$$\frac{V_{o2}}{V_i} = -\frac{1}{1+sCR^2/R_1(1+sCR_1)}$$

$$\frac{V_{o1}}{V_i} = \frac{\left(\frac{R}{R_1}+sCR\right)}{1+sCR^2/R_1(1+sCR_1)}$$

$$\frac{-(V_{o1}+1/s)}{1/sC} - \frac{V_{o2}}{R} = 0$$

$$V_{o1}(sC) + V_{o2}(1/R) = -C$$

$$\frac{V_{o1}}{R} - V_{o2}.sC \Rightarrow V_{o2} = V_{o1}(1/sCR)$$

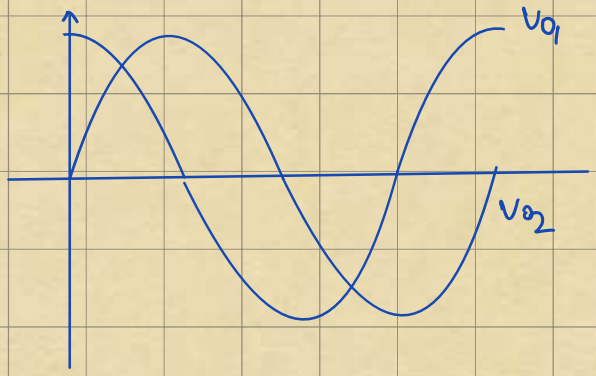
$$V_{o1}(sC) + V_{o1}(1/sCR^2) = -C$$

$$\Rightarrow V_{o1}\left(1+\frac{1}{s^2CR^2}\right) = -sC^2R^2$$

$$V_{o1} = \frac{-sC^2R^2}{1+s^2C^2R^2} = \frac{s}{s^2+\omega^2}$$

$$V_{o1} = \cos(\omega_0 t)u(t)$$

$$V_{o2} = \sin(\omega_0 t)u(t)$$



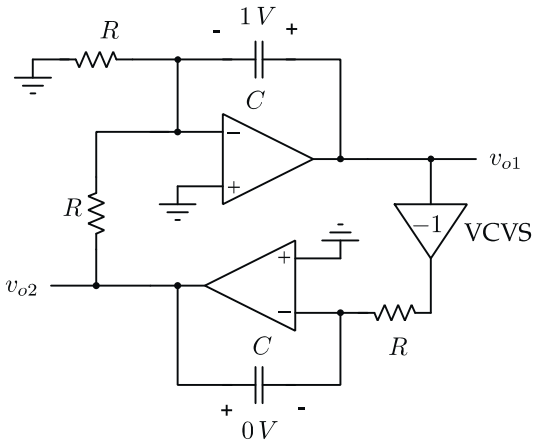


Figure 5: Circuit for Problem 5.

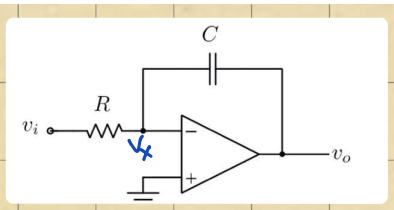
Problem 9

Consider the integrator of Fig. 1. The opamp is not ideal, but has a frequency dependent gain determined by GB/s, where GB denotes its gain-bandwidth product. Determine the integrator's transfer function, when a nonideal opamp is used.

Problem 10

Use the results of Problem 9 to evaluate the transfer function of the circuit of Fig. 4 when the opamps have a finite gain-bandwidth product. The VCVS can be assumed to be ideal.

Consider the integrator of Fig. 1. The opamp is not ideal, but has a frequency dependent gain determined by GB/s, where GB denotes its gain-bandwidth product. Determine the integrator's transfer function, when a nonideal opamp is used.



$$-V_x\left(\frac{GB}{s}\right) = V_o$$

$$V_x = \frac{V_o R + V_i/sC}{R + 1/sC} \Rightarrow \frac{V_i + V_o(sRC)}{1+sRC} = -\frac{sV_o}{GB}$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{-1}{s(RC + 1/GB + sRC/GB)}$$

EE2019 – Analog Systems and Lab.

Tutorial #5

Qadeer Ahmad Khan & Saurabh Saxena

Question-1

A filter transfer function is given by

$$H(s) = \frac{10\left(1 + \frac{s^2}{\omega_p^2}\right)}{1 + \frac{s}{\omega_p Q_p} + \frac{s^2}{\omega_p^2}}$$

Here, $\omega_p = 10^8$ and $Q_p = \frac{1}{3}$.

a) Draw s-domain block diagram of the above filter using integrators and gain blocks.
b) Realize the above filter transfer function using ideal opamps, resistors, and capacitors.
c) If 10pF capacitors are used for all integrators, find all resistor values used in your design.
d) Draw Bode magnitude and phase plots for the filter transfer function.

Question-2

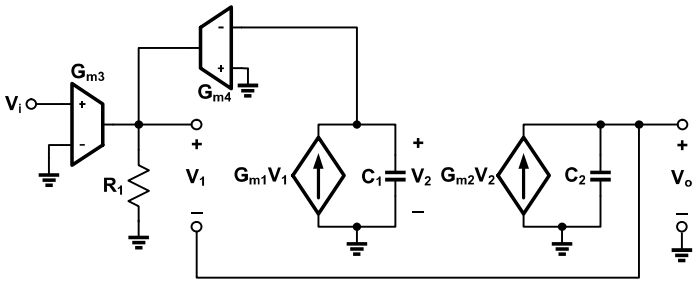
The magnitude of a filter transfer function is given by

$$|H(j\omega)|^2 = \frac{4}{1 + \left(\frac{\omega}{\omega_p}\right)^6}$$

Here, $\omega_p = 10^8$.

a) Find transfer function for the filter in s-domain with magnitude response as given above. The filter should be stable.
b) Draw s-domain block diagram of the above filter using integrators and gain blocks.
c) Realize the above filter transfer function using ideal opamps, resistors, and capacitors.
d) If 10pF capacitors are used for all integrators, find all resistor values used in your design.
e) Draw Bode magnitude and phase plots for the filter transfer function.

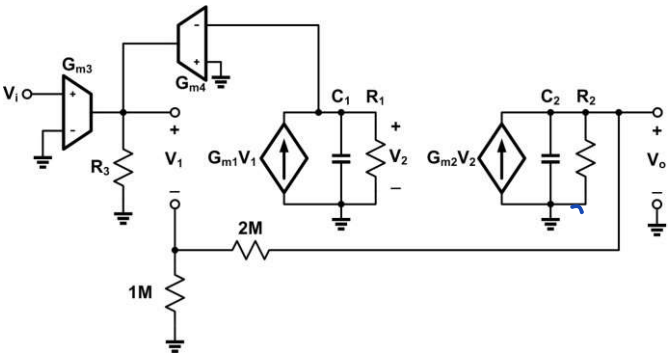
Question-3



For the closed loop feedback system shown above G_{m1} = 4 mA/V, G_{m2} = 25mA/V, G_{m3} = 1 mA/V, R_1 = 1kΩ, C_1 = 10 pF, C_2 = 100 pF,

a) Find loop gain and phase margin when G_{m4} = 0.
b) Find G_{m4} such that the closed loop $\frac{V_o(s)}{V_i(s)}$ is a second-order transfer function with Butterworth (maximally flat) response in signal band.
c) For G_{m4} value found in part (b), find loop gain and plot Bode magnitude and phase plots for the loop gain. Also, find the phase margin for loop.

Question-4



For the closed loop feedback system shown above G_{m1} = 25 mA/V, G_{m2} = 40 mA/V, G_{m3} = 1 mA/V, R_1 = 1kΩ , R_2 = 100Ω, R_3 = 1kΩ, C_1 = 100 pF, C_2 = 1 nF,

a) Find loop gain and phase margin when G_{m4} = 0.
b) Find G_{m4} such that the closed loop $\frac{V_o(s)}{V_i(s)}$ is a second-order transfer function with Butterworth (maximally flat) response in signal band.
c) For G_{m4} value found in part (b), find loop gain and plot Bode magnitude and phase plots for the loop gain. Also, find the phase margin for loop.

The magnitude of a filter transfer function is given by

$$|H(j\omega)|^2 = \frac{4}{1+\left(\frac{\omega}{\omega_p}\right)^6}$$

Here, $\omega_p = 10^8$.

a) Find transfer function for the filter in s-domain with magnitude response as given above. The filter should be stable.
b) Draw s-domain block diagram of the above filter using integrators and gain blocks.
c) Realize the above filter transfer function using ideal opamps, resistors, and capacitors.
d) If 10pF capacitors are used for all integrators, find all resistor values used in your design.
e) Draw Bode magnitude and phase plots for the filter transfer function.

$$H(s)H(-s) = \frac{4}{1-(s/\omega_p)^6}$$

poles: $s^6 = -\omega_p^6 = \omega_p^6 e^{j\pi(2k+1)}$

$$p_1 = \omega_p\left(-\frac{1+\sqrt{3}j}{2}\right)$$

$$p_2 = -\omega_p$$

$$p_3 = \omega_p\left(\frac{-1+\sqrt{3}j}{2}\right)$$

a) $$H(s) = \frac{2}{1+2\times10^{-8}s+2\times10^{-16}s^2+10^{-24}s^3} = \frac{Y(s)}{X(s)}$$

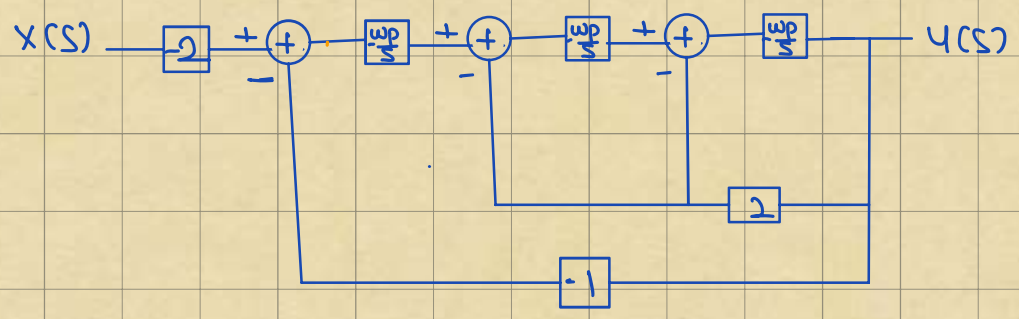
b) $$Y(s)+2\times10^{-8}sY(s)+2\times10^{-16}Y(s)\cdot s^2+10^{-24}Y(s)s^3 = 2X(s)$$

$$10^{24}s^{-3}Y(s)+2\times10^{16}s^{-2}Y(s)+2\times10^{8}s^{-1}Y(s)+Y(s) = 2\times10^{24}s^{-3}X(s)$$

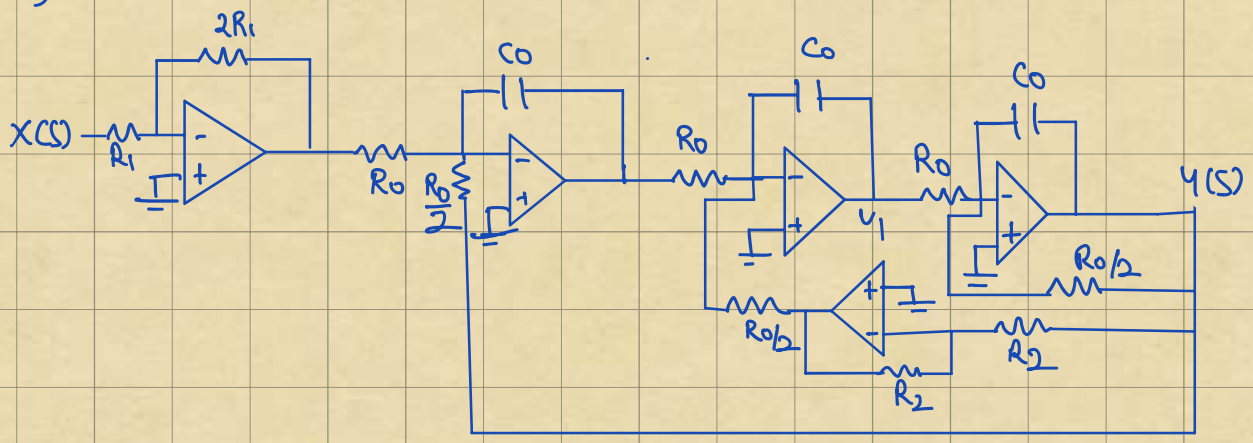
$$Y(s) = -\frac{10^{24}}{s^3}Y(s)-\frac{2\times10^{16}}{s^2}Y(s)-\frac{2\times10^{8}}{s}Y(s)+\frac{2\times10^{24}}{s^3}X(s)$$

$$Y(s) = -\frac{10^{24}}{s^3}Y(s)-\frac{2\times10^{16}}{s^2}Y(s)-\frac{2\times10^{8}}{s}Y(s)+\frac{2\times10^{24}}{s^3}X(s)$$

$$Y(s) = (\omega_p/s)^3(+Y(s)-2X(s))+(\omega_p/s)^2(-2Y(s))+(\omega_p/s)(+2Y(s))$$



c)



d) $\frac{1}{R_0 C_0} = \omega_p \Rightarrow 10^8 = \frac{10^{11}}{R_0} \Rightarrow R_0 = 1\ K\Omega$

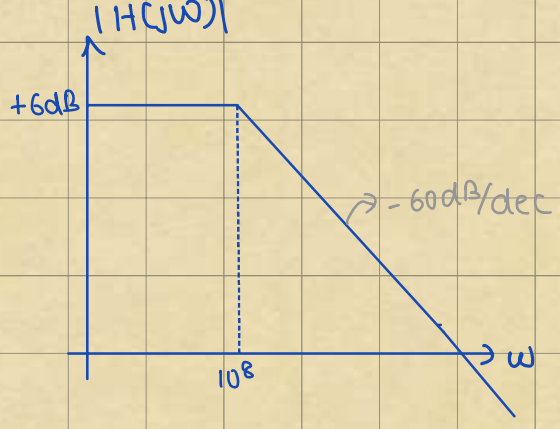
$R_1 = 5K\Omega \quad R_2 = 5K\Omega$

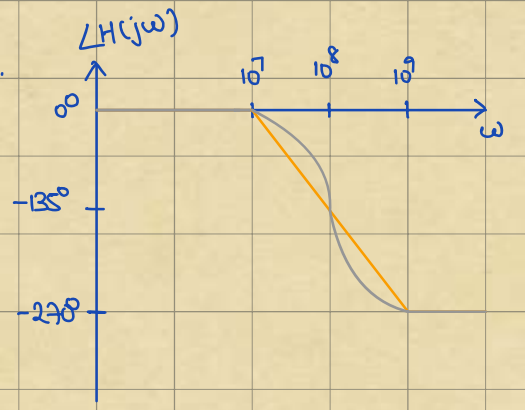
e) $H(S) = \frac{2}{(1+S/10^8)(1+S/10^8+S^2/10^{16})}$

first order pole: 10^8

2nd order system: $\omega_n = 10^8,\ \zeta = 0.5$

DC gain $= |H(0)| = 2 = 6dB$





A filter transfer function is given by

$$H(s) = \frac{10\left(1 + \frac{s^2}{\omega_p^2}\right)}{1 + \frac{s}{\omega_p Q_p} + \frac{s^2}{\omega_p^2}} = \frac{Y(s)}{X(s)}$$

Here, $\omega_p = 10^8$ and $Q_p = \frac{1}{3}$.

a) Draw s-domain block diagram of the above filter using integrators and gain blocks.
b) Realize the above filter transfer function using ideal opamps, resistors, and capacitors.
c) If 10pF capacitors are used for all integrators, find all resistor values used in your design.
d) Draw Bode magnitude and phase plots for the filter transfer function.

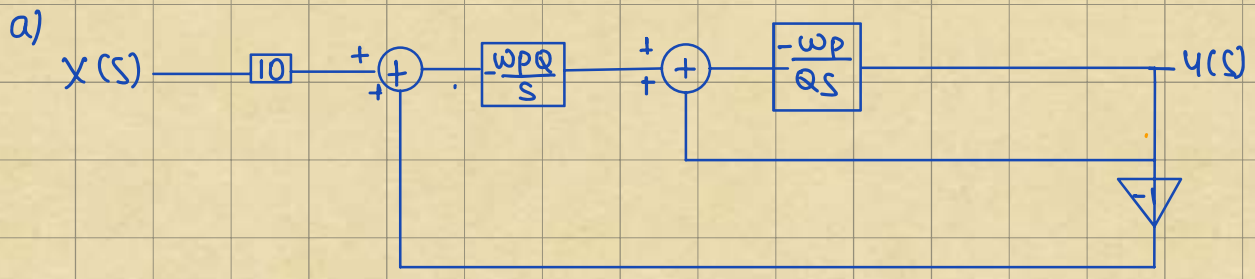
$$(s/\omega_p)^2 Y(s) + (s/\omega_p)(Y(s)/Q) + Y(s) = 10X(s) + (s/\omega_p)^2 (10X(s))$$

$$\Rightarrow Y(s) + \left(\frac{\omega_p}{s}\right)\left(\frac{Y(s)}{Q}\right) + \left(\frac{\omega_p}{s}\right)^2 Y(s) = 10X(s) + \left(\frac{\omega_p}{s}\right)^2 (10X(s))$$

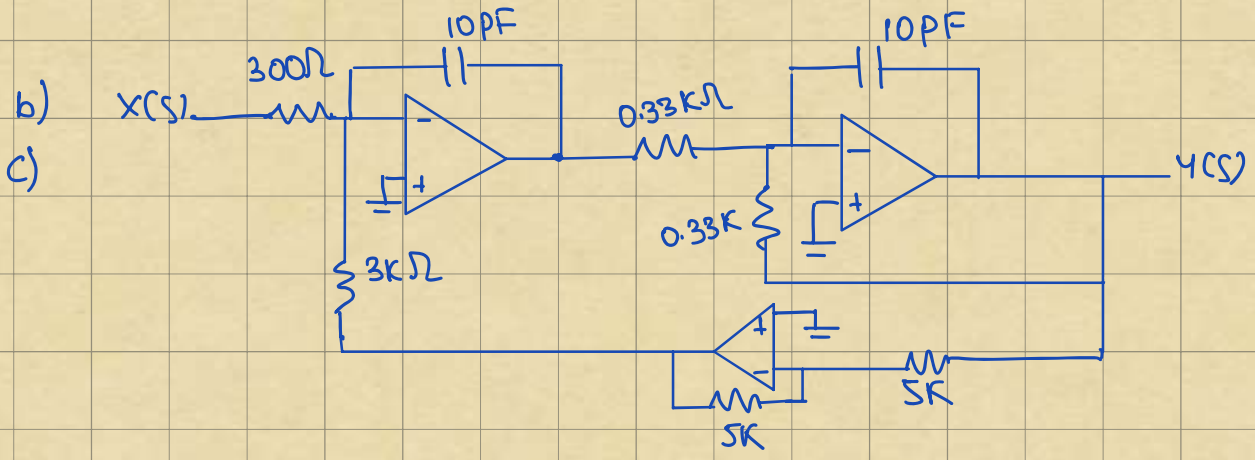
$$Y(s) = \left(\frac{\omega_p}{s}\right)^2 [10X(s) - Y(s)] + \frac{\omega_p}{s}\left[-Y(s)\frac{1}{Q}\right] + 10X(s)$$

$$= \left[[10X(s) - Y(s)]\left(\frac{\omega_p Q}{s}\right) + Y(s)\left(\frac{-\omega_p}{Qs}\right)\right] + 10X(s)$$

a)



b)
c)

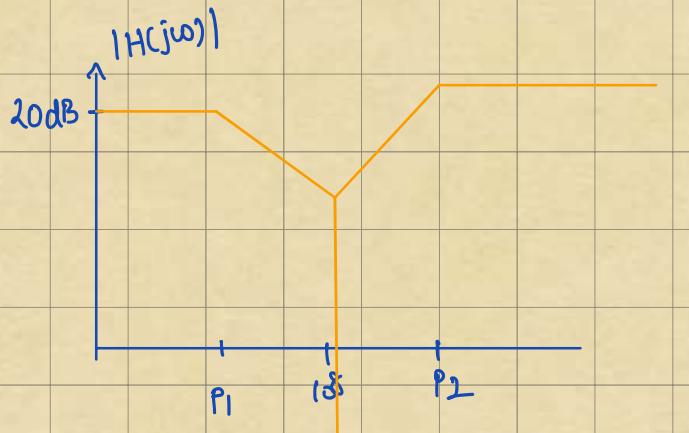


d)

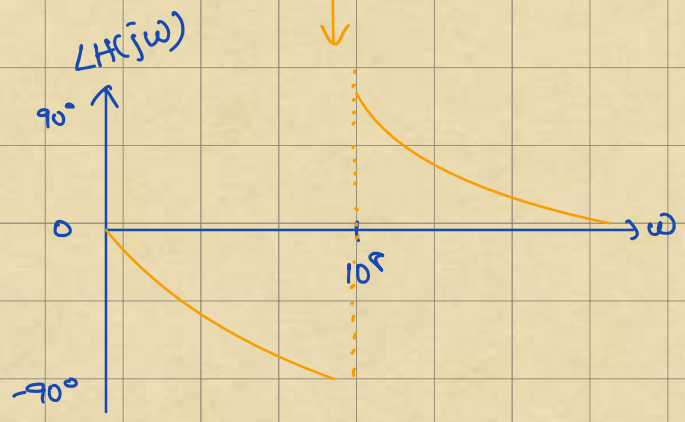
$$H(s) = \frac{10(s^2 + 10^{16})}{s^2 + 3\times10^8 s + 10^{16}}$$

zeros: $s = \pm j10^8$

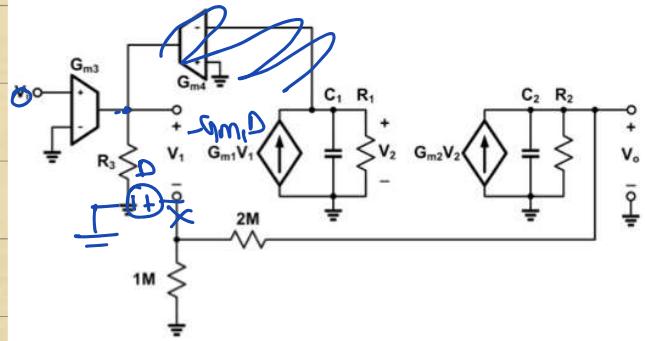
poles: 0.382×10^8, 2.618×10^8 (p_1, p_2)



$$\phi = -\tan^{-1}\left(\frac{3\times10^8\omega}{10^{16} - \omega^2}\right)$$



Question-4



For the closed loop feedback system shown above G_{m1} = 25 mA/V, G_{m2} = 40 mA/V, G_{m3} = 1 mA/V, R_1 = 1kΩ , R_2 = 100Ω, R_3 = 1kΩ, C_1 = 100 pF, C_2 = 1 nF,

a) Find loop gain and phase margin when G_{m4} = 0.

b) Find G_{m4} such that the closed loop $\frac{V_o(s)}{V_i(s)}$ is a second-order transfer function with Butterworth (maximally flat) response in signal band.

c) For G_{m4} value found in part (b), find loop gain and plot Bode magnitude and phase plots for the loop gain. Also, find the phase margin for loop.

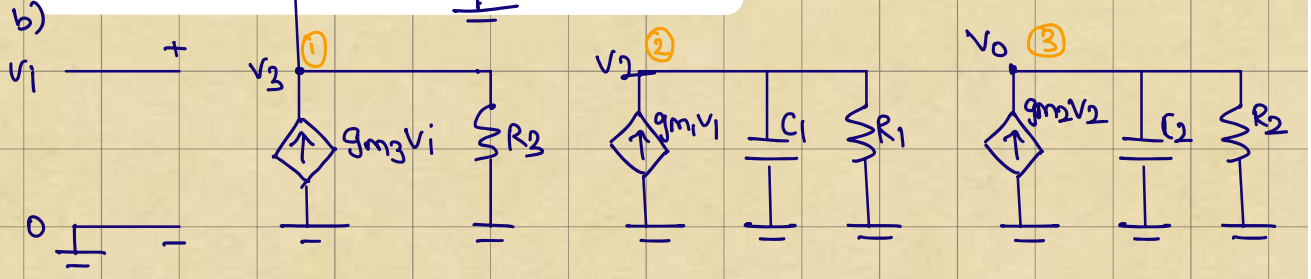
a) $\frac{1}{3}\cdot\frac{g_{m1}R_1}{1+sC_1R_1}\cdot\frac{g_{m2}R_2}{1+sC_2R_2}$

$\frac{1}{3}\cdot\frac{g_{m1}g_{m2}}{C_1C_2}\cdot\frac{1}{(s+1/R_1C_1)(s+1/R_2C_2)}$

$V_1 = V_3 - \frac{V_o}{3}$ $\quad V_3 = V_1 + \frac{V_o}{3}$

$G_2 = 10^{-2} + \frac{1}{3}\times 10^{-6}$

b)



$g_{m3}V_i = g_{m4}V_2 + V_3G_3 \Rightarrow g_{m3}V_i = \frac{g_{m4}V_o(sC_2+G_2)}{g_{m2}} + \frac{V_oG_3}{3} + \frac{G_3V_o(sC_1+G_1)(sC_2+G_2)}{g_{m1}g_{m2}}$

$g_{m1}V_1 = V_2(sC_1+G_1)$

$g_{m2}V_2 = V_o(sC_2+G_2)$

$\Rightarrow \frac{V_o}{V_i} = \frac{3g_{m3}g_{m1}g_{m2}}{3g_{m1}g_{m4}(sC_2+G_2) + g_{m1}g_{m2}G_3 + 3G_3(sC_1+G_1)(sC_2+G_2)}$

$= \frac{3g_{m1}g_{m2}g_{m3}}{s^2(3C_1C_2G_3) + s(3g_{m1}g_{m4}C_2 + 3G_3(C_1G_2+C_2G_1)) + 3g_{m1}g_{m4}G_2 + g_{m1}g_{m2}G_3 + 3G_3G_1G_2}$

$$3C_1C_2G_3 = 3\times10^{-10}\times10^{-9}\times10^{-3} = 3\times10^{-22}$$

$$g_{m1}g_{m2}G_3 = 25\times10^{-3}\times40\times10^{-3}\times10^{-3} = 10^{-6}$$

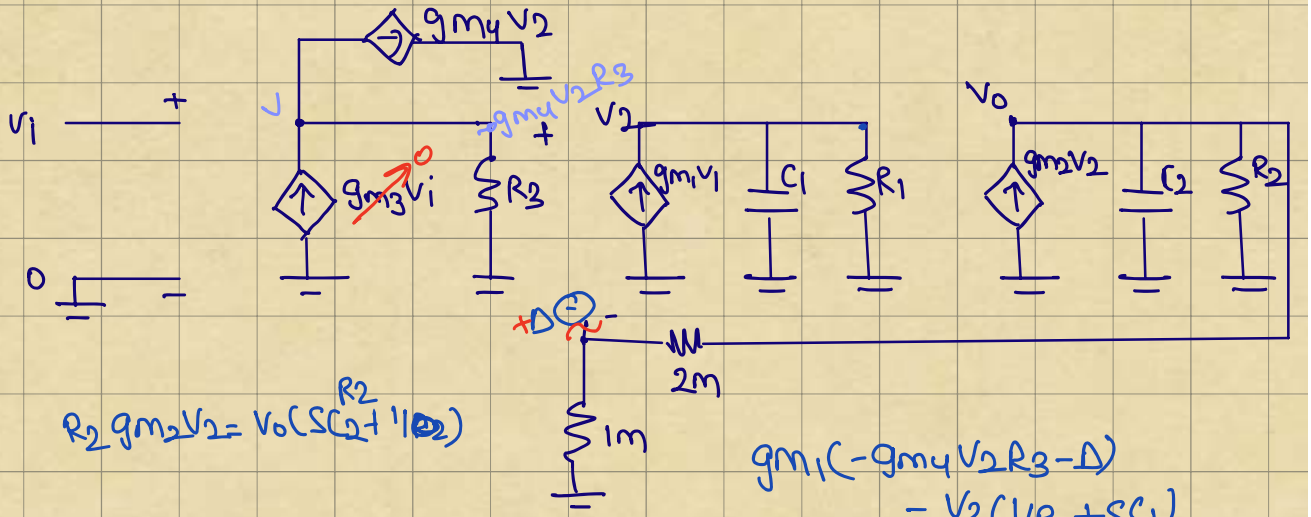
$$3G_1G_2G_3 = 3\times10^{-3}\times10^{-3}\times10^{-2} = 3\times10^{-8}$$

$$3G_3(C_1G_2+C_2G_1) = 3\times10^{-3}[2\times10^{-12}] = 6\times10^{-15}$$

$$\frac{V_o}{V_i} = \frac{3\times10^{-6}}{(3\times10^{-22})s^2 + (75\times10^{-12}g_{m4} + 6\times10^{-15})s + (75\times10^{-5}g_{m4} + 103\times10^{-8})}$$

$$D(s) = as^2+bs+c \Rightarrow b^2 = 2ac \Rightarrow g_{m4} = 0.284\ mA/V$$

$$\frac{V_o}{V_i} = \frac{3\times10^{-6}}{3\times10^{-22}s^2 + 2.73\times10^{-14}s + 12.43\times10^{-7}}$$



$$R_2 g_{m2}V_2 = V_o(sC_2 + 1/R_2)$$

$$g_{m1}(-g_{m4}V_2R_3 - D) = V_2(1/R_1 + sC_1)$$

$$-g_{m1}D = V_2(1/R_1 + g_{m1}g_{m4}R_3 + sC_1)$$

$$V_2 = \frac{-g_{m1}R_1D}{1+g_{m1}g_{m4}R_1R_3+sC_1R_1}$$

$$V_o = V_2 \cdot \frac{g_{m_2} R_2}{1 + sC_2R_2}$$

$$LG = \frac{1}{3} \frac{g_{m_1} R_1}{(1 + g_{m_1} g_{m_4} R_1 R_3 + sC_1R_1)} \frac{g_{m_2} R_2}{(1 + sC_2R_2)}$$

$$= \frac{1}{3} \frac{g_{m_1} g_{m_2} R_1 R_2}{s^2(C_1C_2R_1R_2) + s(C_2R_2 + C_1R_1 + C_2R_2 g_{m_1} g_{m_4} R_1R_3) + 1 + g_{m_1} g_{m_4} R_1 R_3}$$

$$= \frac{1}{3} \frac{100}{10^{-14} s^2 + 9.1 \times 10^{-7} s + 8.1}$$

$$|LG| = 1 \Rightarrow \frac{(100/3)^2}{(8.1 - \omega^2 \cdot 10^{-14})^2 + 9.1^2 \times 10^{-14} \omega^2} = 1 \Rightarrow \omega = 3.62 \times 10^7 \text{ rad/s}$$

$$p_1 = -10^7, \quad p_2 = -8.1 \times 10^7$$

$$PM = 180 + \tan^{-1}\left(\frac{-1}{3.62}\right) + \tan^{-1}\left(\frac{-8.1}{3.62}\right)$$

ANALOG SYSTEMS : PROBLEM SET 6

Problem 1

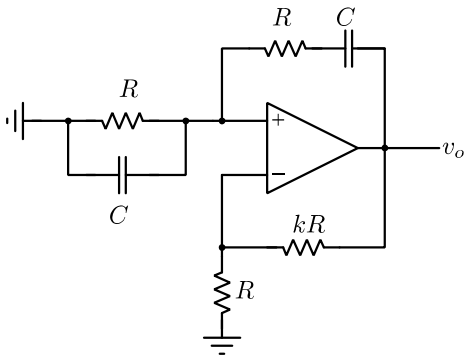


Figure 1: Circuit for Problem 1.

Fig. 1 shows a sinewave oscillator. Determine k so that it just begins to oscillate. All opamps are ideal.

Problem 2

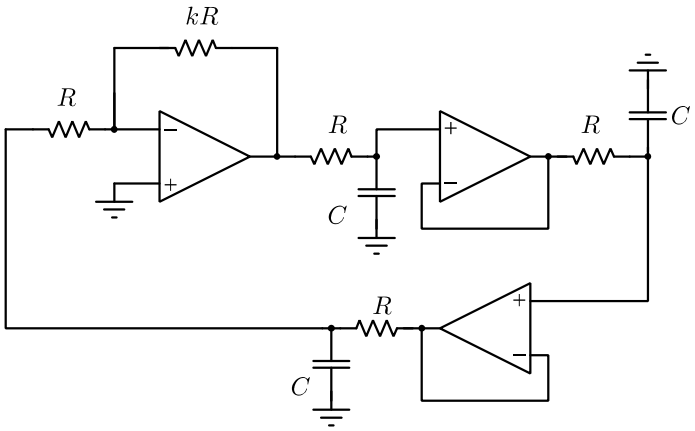


Figure 2: Circuit for Problem 2.

All opamps are ideal in Fig. 2. Determine k so that it just begins to oscillate.

Problem 3

Repeat for the circuit of Fig. 3.

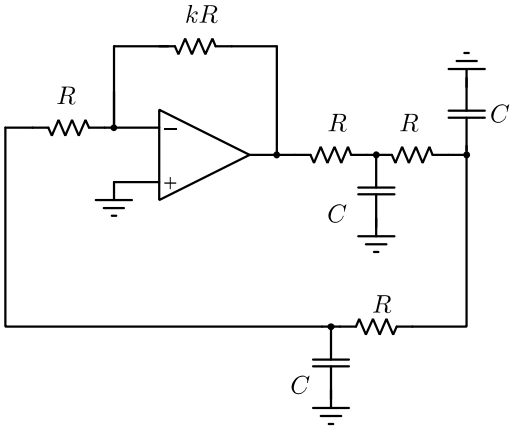


Figure 3: Circuit for Problem 3.

Problem 4

Fig. 4(a) shows an LCR network with a limiting VCCS. The Q of the RLC parallel network can be assumed to be $\gg 1$. The characteristic of the VCCS is shown in Fig. 4(b). The slope of the VCCS is denoted by G_1 and the maximum/minimum current it can source/sink is given by $\pm I_{max}$.

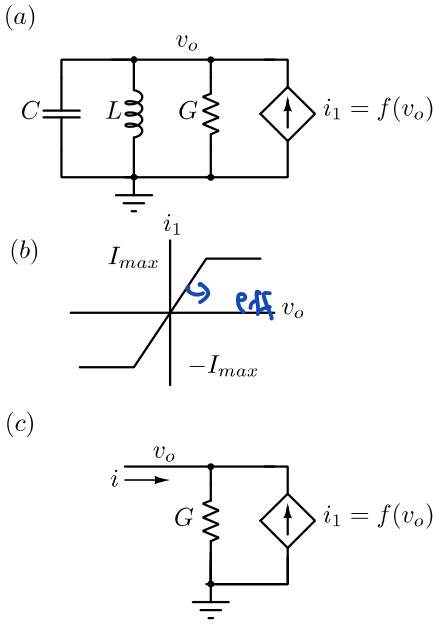


Figure 4: Circuit for Problem 4.

Determine the condition on G_1 for oscillation to start up. Assuming that this condition is satisfied, draw the $i - v_o$ characteristic of the element shown in Fig. 4(c). Assume that $v_o = A\sin(\omega t)$. Determine and plot the amplitude of the fundamental component of the current i as A is varied from 0 to ∞. What will be amplitude of oscillation of the circuit in part (a) of the figure, in steady state? You may assume that v_o is a sinusoid at the fundamental frequency.

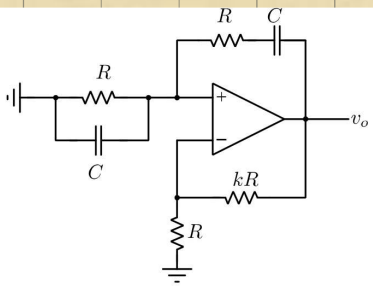


Figure 1: Circuit for Problem 1.

Fig. 1 shows a sinewave oscillator. Determine k so that it just begins to oscillate. All opamps are ideal.

$$V_+ = \frac{V_o}{\frac{G+SC}{\frac{1}{G+SC} + R + \frac{1}{SC}}} = \frac{V_o}{3 + 1/SCR + SCR}$$

$$V_- = \frac{V_o}{K+1} \Rightarrow K+1 = 3 + \frac{1}{SCR} + SCR$$

$$K+1 = 3 - \frac{j}{RC\omega} + j\omega CR$$

$$K = 2 \; ; \; \omega = \frac{1}{RC}$$